Drift-Aware Eigenvalue Scaling for Gradient Descent Under Changing Curvature: A Reproducible Simulation Study

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Abstract

A scalar step size that is effective for one objective can become too aggressive when the objective's curvature changes. We evaluate a lightweight drift-aware heuristic that estimates the largest Hessian eigenvalue online with a short power iteration and adapts a scalar gradient-descent step with a safety factor and an upward-responsive exponential tracker. The primary experiment is a two-dimensional strongly convex quadratic whose eigenvalues, eigendirections, and minimizer vary over time. The supplied implementation's default seed-4 result is reproduced exactly: plain gradient descent finishes 260 updates with gap 0.0631940560, frozen initial eigen scaling crosses the implementation's numerical threshold at update 42, and drift-aware scaling finishes with gap 0.0355166284. A new matched study covers three drift regimes, six methods, and 30 independent seeds per regime. Across all regimes, frozen scaling fails in all 30 runs, while plain, drift-aware, oracle, and both tracker ablations complete all runs. On the source-matched moderate regime, the drift-aware method has mean final gap 0.0355166 (95% Student-t CI $[0.0355166, 0.0355166]$) and mean absolute filtered-eigenvalue relative error 0.00626; these narrow intervals reflect a deterministic two-dimensional schedule in which seed variation enters only through the initial point. A separate 360-run classification study uses the repository's materially different HVP-norm rule on nominal covariate-drift streams. The tuned eigen-norm method is strongest in the held-out accuracy and balanced-accuracy summaries, but this evidence remains synthetic and method-specific. The results support a controlled, reproducible demonstration of a scalar step-size heuristic, not a general optimizer, universal superiority claim, or non-stationary stability theorem.

Keywords: gradient descent; changing curvature; eigenvalue estimation; power iteration; adaptive step size; numerical stability; prequential evaluation

1 Introduction

Gradient methods are sensitive to curvature. For a fixed smooth quadratic, the largest Hessian eigenvalue determines the most restrictive direction for a constant step. A step calibrated when curvature is modest can later become unstable if the largest eigenvalue grows, whereas a deliberately small step can remain safe but make slow progress. The classical fixed-objective analysis makes this dependence explicit [7].

This study examines a narrow experimental question: under controlled changes in a strongly convex quadratic objective, can an online estimate of the largest curvature improve the stability–progress trade-off relative to a conservative fixed step and a scale frozen at initialization? The method is intentionally modest. It performs a few matrix–vector power iterations, computes a Rayleigh quotient, filters the estimate, and applies a scalar step-size rule. It does not form

a matrix preconditioner, change coordinates, or claim a convergence theorem for arbitrary time-varying objectives.

The contribution is therefore empirical and reproducibility-oriented. We preserve the supplied implementation and its original default comparison, add independent repeated-seed experiments and explicit drift regimes, quantify estimator error and safety ratios, compare an exact-eigenvalue oracle and tracker ablations, and validate the repository's separate streaming-classification scripts under a held-out, prequential protocol. All reported tables and figures are generated from machine-readable run-level outputs.

2 Related work and scope

Gradient-method stability for fixed objectives is classical [7]. Scalar adaptive step-length rules also have a long history, including the two-point Barzilai–Borwein method [1]. AdaGrad and Adam adapt updates using accumulated gradient geometry or moment estimates rather than explicitly tracking the largest Hessian eigenvalue [2, 3]. A few matrix–vector power iterations are a standard spectral approximation device; for numerical-linear-algebra context see Martinsson and Tropp [4].

Diagonal scaling and equilibration are related but distinct. They seek diagonal matrices that transform rows and columns or improve a system's conditioning, including in ADMM and graph-projection splitting [8, 6]. The method studied here produces a scalar step-size scale from a spectral estimate; it is not diagonal preconditioning. Finally, dynamic-environment optimization studies tracking and regret under moving objectives [5], but those results do not establish a guarantee for this particular power-iteration tracker.

The verified literature does not justify claiming that the exact combination used here is new, globally convergent, or uniformly better than other adaptive rules. We use the narrower description "drift-aware eigenvalue step-size heuristic."

3 Problem formulation

At update t, let

$$f_t(x) = \frac{1}{2}(x - x_t^\star)^\top A_t (x - x_t^\star), \qquad x, x_t^\star \in \mathbb{R}^2, \tag{1}$$

where A_t is symmetric positive definite. The gradient update is

$$\nabla f_t(x) = A_t(x - x_t^\star), \qquad x_{t+1} = x_t - \eta_t \nabla f_t(x_t). \tag{2}$$

For one fixed matrix A, a positive constant step is linearly stable when

$$0 < \eta < \frac{2}{\lambda_{\max}(A)}. \tag{3}$$

This is a pointwise fixed-objective condition. It is not a theorem for the complete experiment because A_t changes, the target $x_t^\star$ moves, the eigendirection estimate is approximate, and the selected step can lag the current spectrum.

The source schedule uses

$$\lambda_{\min}(t) = 0.65 + 0.18\sin(\phi_t + 0.75), \tag{4}$$

$$\lambda_{\max}(t) = 8 + 13\frac{1 + \sin(\phi_t - 0.85)}{2} + 11\frac{t}{T-1}, \tag{5}$$

$$\phi_t = \frac{2\pi t}{T-1}, \tag{6}$$

with a rotating eigenbasis and a smooth moving minimizer. The new low and high regimes retain the same construction but use respectively $(6, 4)$ and $(18, 18)$ for the periodic-amplitude and linear-drift coefficients, together with stronger or weaker rotation and target-motion schedules. The exact configuration is in `quadratic/study_config.json`.

4 Methods

4.1 Compared update rules

Plain gradient descent uses $\eta_t = 0.035$. Frozen eigen scaling computes $\lambda_{\max}(A_0)$ once and uses $\eta_t = \gamma/\lambda_{\max}(A_0)$ with $\gamma = 1.78$. The practical drift-aware method initializes $v_0 = (0.83, 0.41)/\|(0.83, 0.41)\|$ and performs seven power iterations per update. After normalization, it computes

$$r_t = v_t^\top A_t v_t, \qquad d_t = \frac{|r_t - r_{t-1}|}{\max(|r_{t-1}|, 10^{-12})}. \tag{7}$$

The smoothing coefficient is

$$a_t = \mathrm{clip}(a_0 + \delta d_t, a_0, a_{\max}), \tag{8}$$

with $(a_0, a_{\max}, \delta) = (0.16, 0.82, 7)$. The filtered estimate is

$$\widehat{\lambda}_t = (1 - a_t)\widehat{\lambda}_{t-1} + a_t r_t. \tag{9}$$

When $r_t > \widehat{\lambda}_{t-1}$, the source code replaces the filtered value by at least r_t. The step is

$$\eta_t = \frac{\gamma}{(1 + s)\widehat{\lambda}_t}, \qquad (\gamma, s) = (1.78, 0.10). \tag{10}$$

We compare two idealized or diagnostic variants. The oracle uses the true current $\lambda_{\max}(A_t)$ in the same safety-adjusted rule. The fixed-smoothing ablation keeps $a_t = a_0$ rather than responding to spectral drift but retains upward correction. The no-upward-correction ablation uses drift-responsive smoothing but removes the asymmetric correction. These ablations are not proposed methods; they isolate design choices.

4.2 What the upward correction does not guarantee

The upward correction guarantees only that, at an update where the raw Rayleigh estimate rises above the previous filtered estimate, the stored filtered estimate is no smaller than that raw estimate. It does not guarantee that the Rayleigh estimate equals the true largest eigenvalue, that the tracked value is monotone, or that $\eta_t < 2/\lambda_{\max}(A_t)$: finite power iteration, changing eigendirections, and the moving objective can still produce underestimation or lag. In the study, safety is therefore measured rather than assumed.

5 Experimental design

5.1 Quadratic study

The primary matrix has 260 updates. Each of the three regimes uses the same 30 independent seeds, with matched starting points across methods. The initial point is $(-2.35, 2.05) + 0.18\,\varepsilon$, where $\varepsilon \sim \mathcal{N}(0, I)$ and the seed is recorded. A run is flagged when the state is non-finite or its Euclidean norm exceeds 10^6; this is an implementation threshold, not a mathematical definition of divergence.

For every update we save the true spectrum, raw and filtered estimates, relative errors, drift score, smoothing coefficient, step size, fixed-Hessian stability threshold, ratio $\eta_t/(2/\lambda_{\max}(A_t))$,

objective gap, distance to the moving minimizer, state norm, and failure status. Run-level summaries report final gap, cumulative gap, tracking distance, estimator error, underestimation frequency, safety ratio, runtime, and failure frequency. Confidence intervals are two-sided Student-t intervals across seeds.

5.2 Classification study

The supplied repository contains two 10,000-row CSV streams with 20 features and a binary target. The covariate stream has overall prevalence 0.1659 and the extreme stream 0.1695. Feature dispersion grows by temporal chunk. The source construction includes label noise (`flip_y=0.05`) and chunk-specific affine transforms, so these are nominal drift streams rather than a pure covariate-shift benchmark.

The independent classification harness uses five tuning streams and 30 held-out streams per condition. The first 2,500 examples fit the scaler and warm up the model; the final 7,500 examples are evaluated pre-update in batches of 100. Six candidates per method are selected using tuning balanced accuracy, then lower log loss as a tie-break. The evaluated eigen method intentionally preserves the source's different HVP-norm rule; it is not the quadratic EMA tracker. We report accuracy, unregularized log loss, balanced accuracy, majority-baseline accuracy, runtime, and failures.

6 Results

6.1 Default source comparison

The source result is reproduced exactly by the independent harness. The table is generated from `quadratic/baseline_check.json`.

Method	Final gap	Updates	Status
Plain GD	0.0631941	260	stable
Frozen eigen	3.05895e+13	42	diverged at 42
Drift-aware	0.0355166	260	stable

The single-run reduction from 0.0631940560 to 0.0355166284 is 43.80%. This percentage is descriptive of the supplied seed and configuration only; the repeated study below is the appropriate uncertainty-bearing comparison.

6.2 Repeated quadratic comparison

Table 1 is generated from the 540 run records. Frozen scaling fails in all 30 runs in every regime. Plain GD and all four current-curvature variants complete all runs. The oracle's safety ratio is $1.78/[2(1.10)] = 0.80909$ before estimator error, while the practical tracker may exceed that ratio when its filtered estimate underestimates the true eigenvalue.

On the source-matched moderate regime, drift-aware scaling completes every run with mean final gap 0.0355166 and mean cumulative gap 66.4327, compared with 0.0631046 and 136.7642 for plain GD. The no-upward-correction ablation has a lower final gap in this particular schedule, 0.0343816, but its mean absolute estimator error is larger (0.0200 versus 0.00626) and it underestimates the true eigenvalue in about 58.8% of updates on average. This is why final objective gap alone is not treated as proof that the full tracker is preferable.

6.3 Spectral tracking and ablations

The drift-aware filtered estimate has mean absolute relative error 0.00533, 0.00626, and 0.00590 in the low, moderate, and high regimes, respectively. Its average underestimation frequency is

Table 1: Repeated quadratic study. Intervals are 95% Student-t CIs across 30 seeds. The failure column is the number of implementation-threshold failures over 30 runs.

Regime	Method	Final gap (95% CI)	Failures/runs	Cumulative gap	max $\eta/(2/\lambda_max)$
Low	Plain GD	0.05416 [0.05411, 0.05421]	0/30	137.6	0.2732
Low	Frozen eigen	8.831e+12 [8.363e+12, 9.299e+12]	30/30	4.022e+12	1.395
Low	Drift-aware	0.0003812 [0.0003812, 0.0003812]	0/30	54.99	0.8091
Low	Oracle current eigen	0.0003802 [0.0003802, 0.0003802]	0/30	54.95	0.8091
Moderate	Plain GD	0.0631 [0.06305, 0.06316]	0/30	136.8	0.4458
Moderate	Frozen eigen	3.135e+13 [2.968e+13, 3.302e+13]	30/30	8.336e+12	1.59
Moderate	Drift-aware	0.03552 [0.03552, 0.03552]	0/30	66.43	0.8091
Moderate	Oracle current eigen	0.03557 [0.03557, 0.03557]	0/30	66.33	0.8091
High	Plain GD	0.1507 [0.1506, 0.1507]	0/30	143.9	0.5844
High	Frozen eigen	4e+13 [3.786e+13, 4.214e+13]	30/30	8.549e+12	1.688
High	Drift-aware	0.03428 [0.03428, 0.03428]	0/30	80.45	0.8091
High	Oracle current eigen	0.03415 [0.03415, 0.03415]	0/30	80.37	0.8091

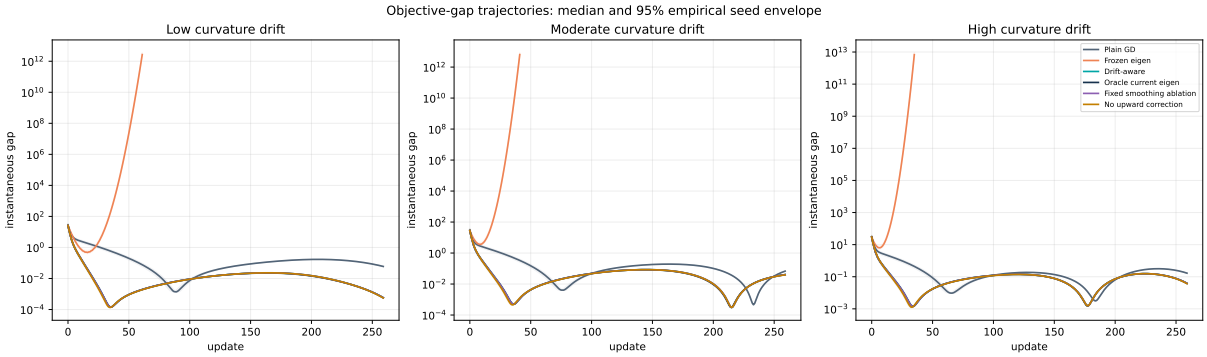


Figure 1: Median objective-gap trajectories with an empirical 95% seed envelope for the three drift regimes. Divergent frozen runs terminate at the implementation threshold.

0.188, 0.219, and 0.212. These are update-level diagnostics, not independent observations for inferential purposes. The oracle has no estimation error by construction.

The no-upward-correction ablation demonstrates that the asymmetric correction is not needed to obtain a low final gap in this one smooth schedule, but it produces more frequent underestimation and a larger average estimation error. The fixed-smoothing ablation is numerically close to the full tracker here, which indicates that the drift-responsive smoothing coefficient is not isolated as the sole cause of the observed behavior. The study therefore supports the mechanism as a controlled heuristic rather than identifying one universally necessary component.

6.4 Classification streams

The classification results are generated from 180 held-out run records, with 30 matched streams per condition and selected hyperparameters fixed from five separate tuning streams. Table 3 reports mean values and run-level 95% CIs.

The eigen-norm rule has the highest mean accuracy in both held-out conditions: 0.8854 on the nominal covariate stream and 0.8593 on the extreme stream. It also has the highest balanced accuracy. Its extreme-stream log loss is higher than Adam's and its confidence interval is wide, so the accuracy ranking is not a complete account of probabilistic performance. All 180 held-out runs completed without a numerical failure. These results cannot be transferred to the quadratic tracker because the classification implementation uses a different estimator and update rule.

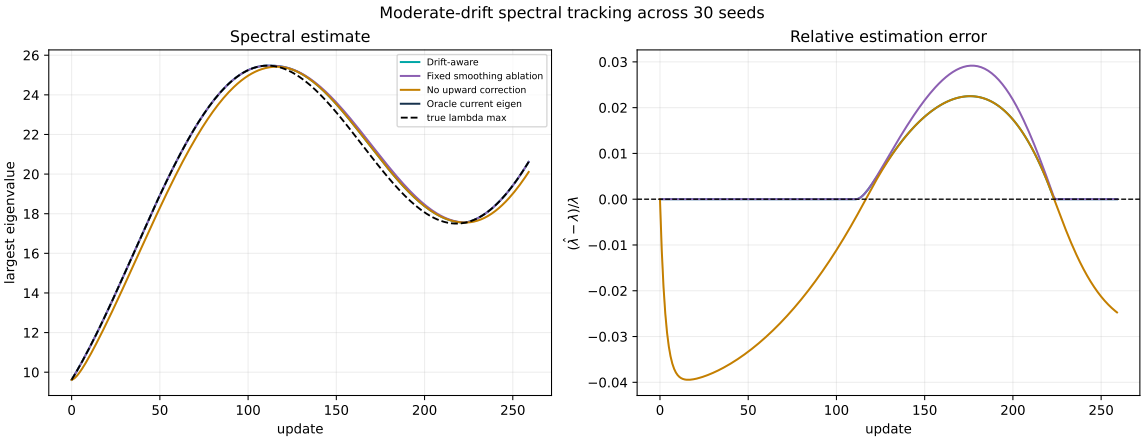


Figure 2: True and estimated largest eigenvalue in the moderate regime. Shaded regions show empirical 95% seed envelopes. The tracker is close on average but is not exact at every update.

Table 2: Moderate-regime tracker ablations generated from the saved run summaries.

Method	Final gap (95% CI)	Mean —relative estimator error—	Underestimate fraction	Max safety rat
Drift-aware	0.035517 [0.035517, 0.035517]	0.006258	0.2192	0.8091
Fixed smoothing ablation	0.035502 [0.035502, 0.035502]	0.007702	0.2192	0.8091
No upward correction	0.034382 [0.034382, 0.034382]	0.02002	0.5885	0.8423

7 Reproducibility and implementation checks

The original Node numerical suite passes all seven tests. The independent Python tests pass four quadratic tests and four classification tests. The quadratic harness records 540 run-level records and 121,200 completed-update records; the classification harness records 360 runs and 36,000 batch records. The source snapshot was hashed before and after the classification study and remained unchanged. The original Python classification scripts were executed with a non-interactive Matplotlib backend, and their logs are retained for provenance.

The package records Python and numerical-library versions in the experiment configuration files, keeps the supplied CSVs unchanged, and writes every result used by this paper to CSV. The exact commands are listed in `README.md` in the output directory.

8 Threats to validity and limitations

1. The quadratic experiment is two-dimensional with a hand-designed smooth schedule. It demonstrates a mechanism but not scalability to large or ill-conditioned machine-learning objectives.

2. The fixed-objective condition $0 < \eta < 2/\lambda_{\max}(A)$ does not imply stability for an arbitrary time-varying process. The adaptive rule remains heuristic.

3. The exact current-eigen oracle is an idealized reference unavailable to the practical method. It separates estimator error from the step-size rule but is not a deployable baseline.

4. The 30 seeds vary only the initial point in the quadratic experiment. The deterministic schedule explains the very small final-gap intervals; they should not be interpreted as evidence of broad stochastic robustness.

5. The ablation results depend on one family of drift profiles. More abrupt changes, faster eigendirection rotation, and faster target motion could alter the ranking.

Table 3: Held-out prequential classification results. The eigen method is the source HVP-norm rule, not the quadratic EMA tracker.

Stream	Method	Accuracy (95% CI)	Log loss (95% CI)	Balanced accuracy (95% CI)	Failures
Covariate	Adam	0.8766 [0.8681, 0.885]	0.3388 [0.3227, 0.355]	0.7075 [0.6868, 0.7283]	0
Covariate	Original eigen-norm	0.8854 [0.8777, 0.8931]	0.3197 [0.3055, 0.3339]	0.7229 [0.7026, 0.7432]	0
Covariate	GD	0.8618 [0.8518, 0.8718]	0.4403 [0.4066, 0.474]	0.695 [0.6744, 0.7157]	0
Extreme	Adam	0.8409 [0.8259, 0.8558]	0.493 [0.4741, 0.5119]	0.7342 [0.7154, 0.7531]	0
Extreme	Original eigen-norm	0.8593 [0.8456, 0.873]	0.7786 [0.7472, 0.81]	0.7645 [0.7482, 0.7809]	0
Extreme	GD	0.8234 [0.806, 0.8408]	0.8599 [0.7487, 0.9711]	0.705 [0.6849, 0.7251]	0

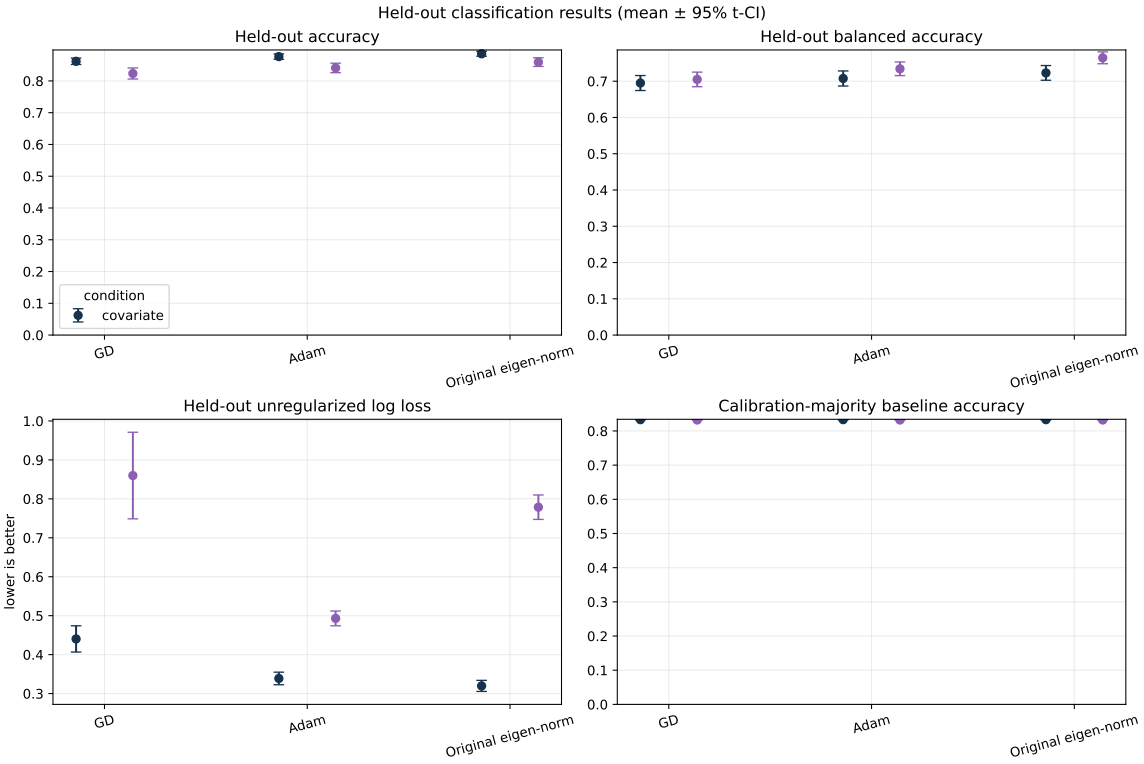


Figure 3: Held-out classification metrics with run-level 95% intervals generated from the classification CSV outputs.

6. The classification streams are synthetic, nominally covariate-drifting, and prequential. Label noise, affine feature transformations, fixed chunk order, and tuning-budget choices constrain interpretation.

7. The classification eigen rule is not the same algorithm as the quadratic tracker. Mixed or favorable classification results cannot validate the quadratic spectral tracker.

8. The supplied historical scripts use an unseeded eigenvector initialization. Their single-run values are preserved as audit records, but the repeated classification study is the reproducible evidence used here.

9. The study did not implement BB or AdaGrad as additional completed baselines; they are included only as verified related work. No claim is made that the method beats them.

9 Conclusion

The supplied default experiment is reproducible: in the source-matched moving quadratic, frozen scaling becomes unsafe while the drift-aware scalar step remains below the implementation's

failure threshold and reaches a lower final gap than plain GD. The repeated study shows that this qualitative pattern persists across the three chosen drift regimes for the frozen-versus-adaptive contrast, but also shows that ablation rankings are schedule-dependent and that estimator error is nonzero. The separate classification study gives favorable held-out accuracy for the source's HVP-norm eigen rule under the evaluated synthetic streams, with materially different algorithmic details and no numerical failures.

The defensible conclusion is therefore narrow: online spectral information can be a useful controlled mechanism for adapting a scalar gradient step under the tested curvature schedules. The evidence does not establish universal superiority, a non-stationary convergence guarantee, or generalization beyond these synthetic experiments. Future work should add high-dimensional objectives, abrupt and adversarial drift, seeded multi-start eigentracking, BB/AdaGrad baselines, and theory or regret analysis appropriate to the time-varying setting.

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